MAHIDOL UNIVERSITY
Faculty of Information and Communication Technology

# Lecture 8
# Transmission Media (Part 2)

September 2026

มหาวิทยาลัยมหิดล
Mahidol University

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# Quotes of the Day



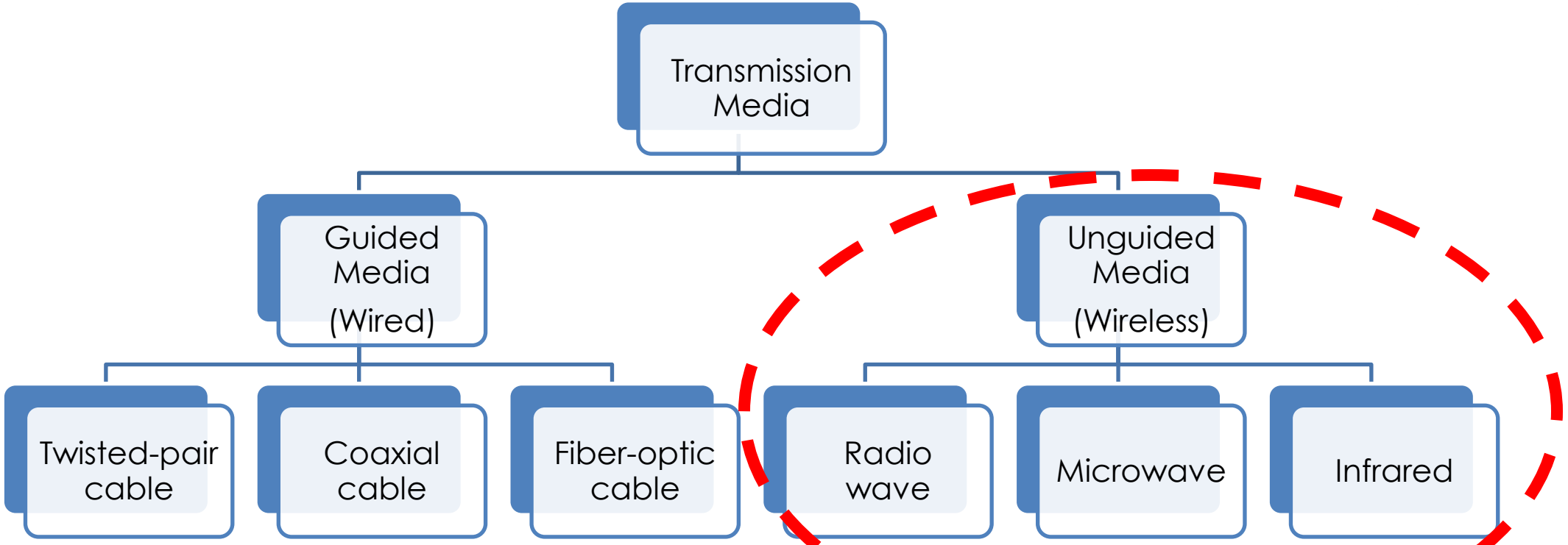

Prince Mahidol Day
24 September



Treat others
the way you want to be treated.

"The true meaning of life is to plant trees under whose shade you do not expect to sit." — Nelson Henderson

# Transmission Media

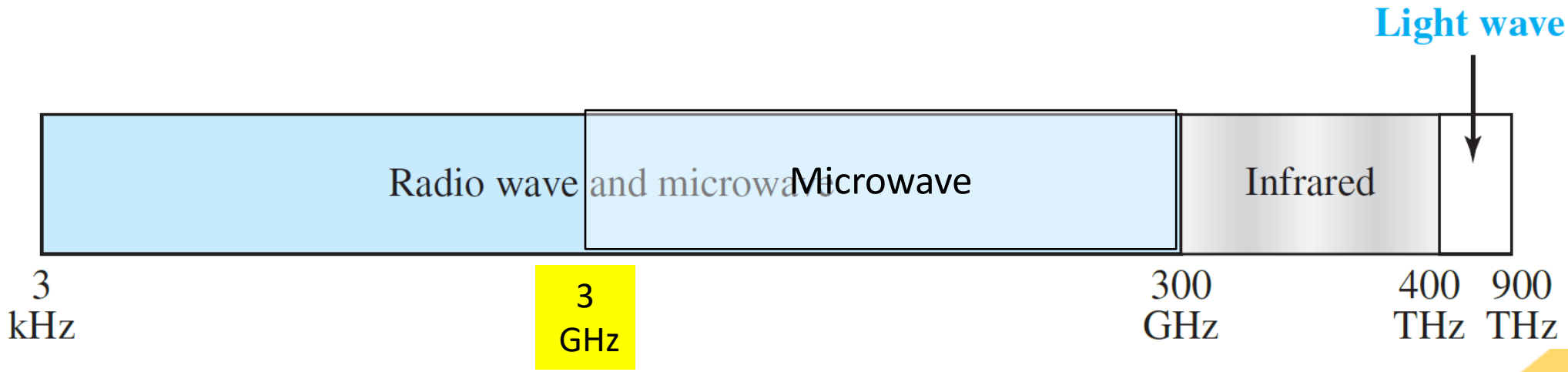



Cable standard: CAT, RG, SMF, MMF diameter of core/ diameter of cladding

Connector standard: RJ, BNC, N, UHF, etc., Connector: MT-RJ, SC, ST, etc.

# Unguided Media

- Unguided medium transport **electromagnetic waves or light (infrared)** without using a physical conductor.
- This type of communication is often referred to as wireless communication.
- Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.



# Regions of the Electromagnetic Spectrum

| | Wavelength (m) | Frequency (Hz) | Energy (J) |
|---|---|---|---|
| **Radio** | $> 1 \times 10^{-1}$ | $< 3 \times 10^{9}$ | $< 2 \times 10^{-24}$ |
| **Microwave** | $1 \times 10^{-3} - 1 \times 10^{-1}$ | $3 \times 10^{9} - 3 \times 10^{11}$ | $2 \times 10^{-24} - 2 \times 10^{-22}$ |
| **Infrared** | $7 \times 10^{-7} - 1 \times 10^{-3}$ | $3 \times 10^{11} - 4 \times 10^{14}$ | $2 \times 10^{-22} - 3 \times 10^{-19}$ |
| **Optical** | $4 \times 10^{-7} - 7 \times 10^{-7}$ | $4 \times 10^{14} - 7.5 \times 10^{14}$ | $3 \times 10^{-19} - 5 \times 10^{-19}$ |
| **UV** | $1 \times 10^{-8} - 4 \times 10^{-7}$ | $7.5 \times 10^{14} - 3 \times 10^{16}$ | $5 \times 10^{-19} - 2 \times 10^{-17}$ |
| **X-ray** | $1 \times 10^{-11} - 1 \times 10^{-8}$ | $3 \times 10^{16} - 3 \times 10^{19}$ | $2 \times 10^{-17} - 2 \times 10^{-14}$ |
| **Gamma-ray** | $< 1 \times 10^{-11}$ | $> 3 \times 10^{19}$ | $> 2 \times 10^{-14}$ |

https://imagine.gsfc.nasa.gov/science/toolbox/spectrum_chart.html

# Radio Frequency Range

| Type of Radiation | Frequency Range (Hz) | Wavelength Range |
|---|---|---|
| gamma-rays | $10^{20}$ - $10^{24}$ | < $10^{-12}$ m |
| x-rays | $10^{17}$ - $10^{20}$ | 1 nm - 1 pm |
| ultraviolet | $10^{15}$ - $10^{17}$ | 400 nm - 1 nm |
| visible | 4 - 7.5*$10^{14}$ | 750 nm - 400 nm |
| near-infrared | 1*$10^{14}$ - 4*$10^{14}$ | 2.5 μm - 750 nm |
| infrared | $10^{13}$ - $10^{14}$ | 25 μm - 2.5 μm |
| microwaves | 3*$10^{11}$ - $10^{13}$ | 1 mm - 25 μm |
| radio waves | < 3*$10^{11}$ | > 1 mm |

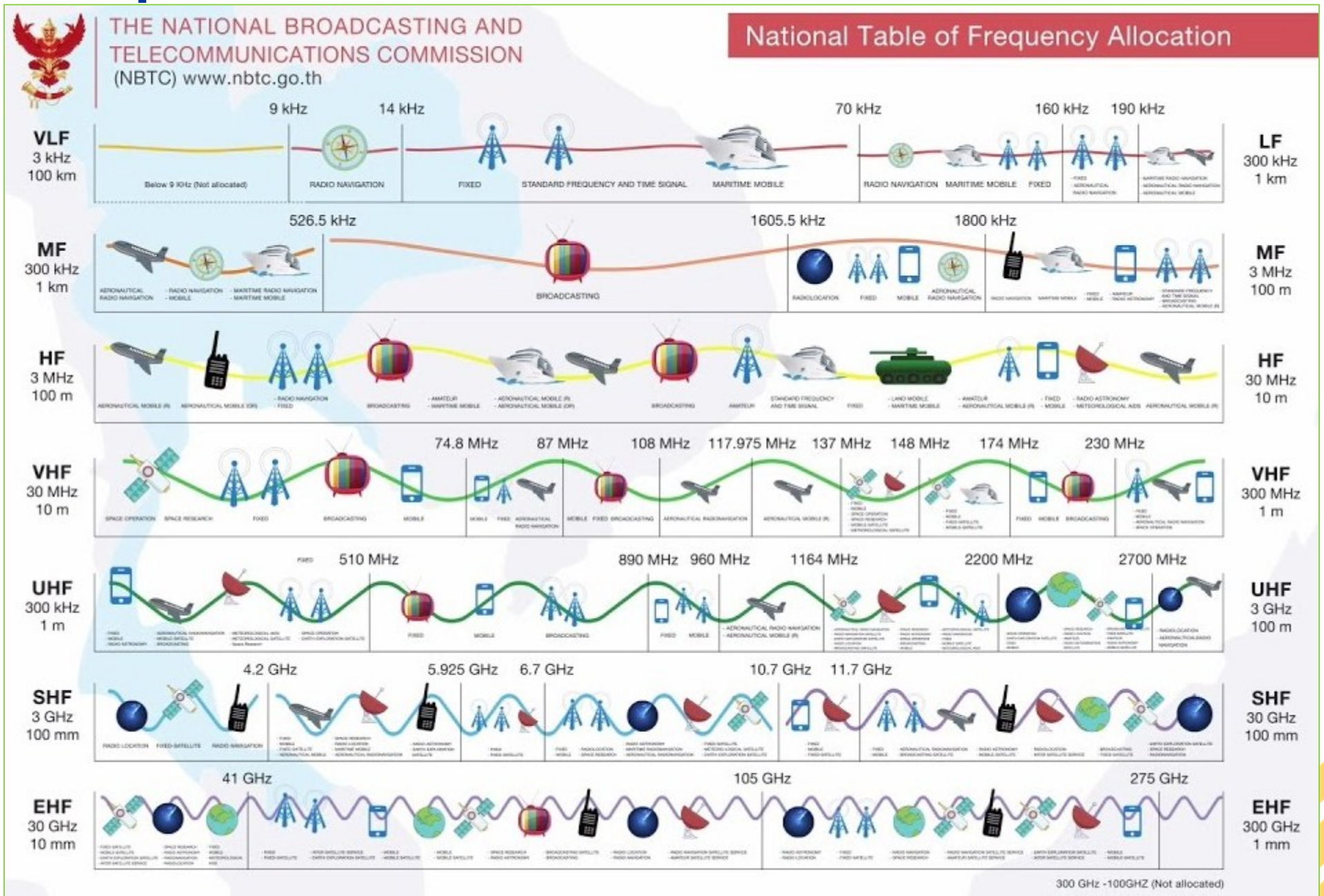
http://labman.phys.utk.edu/phys222core/modules/m6/The%20EM%20spectrum.html

# Frequency Bands

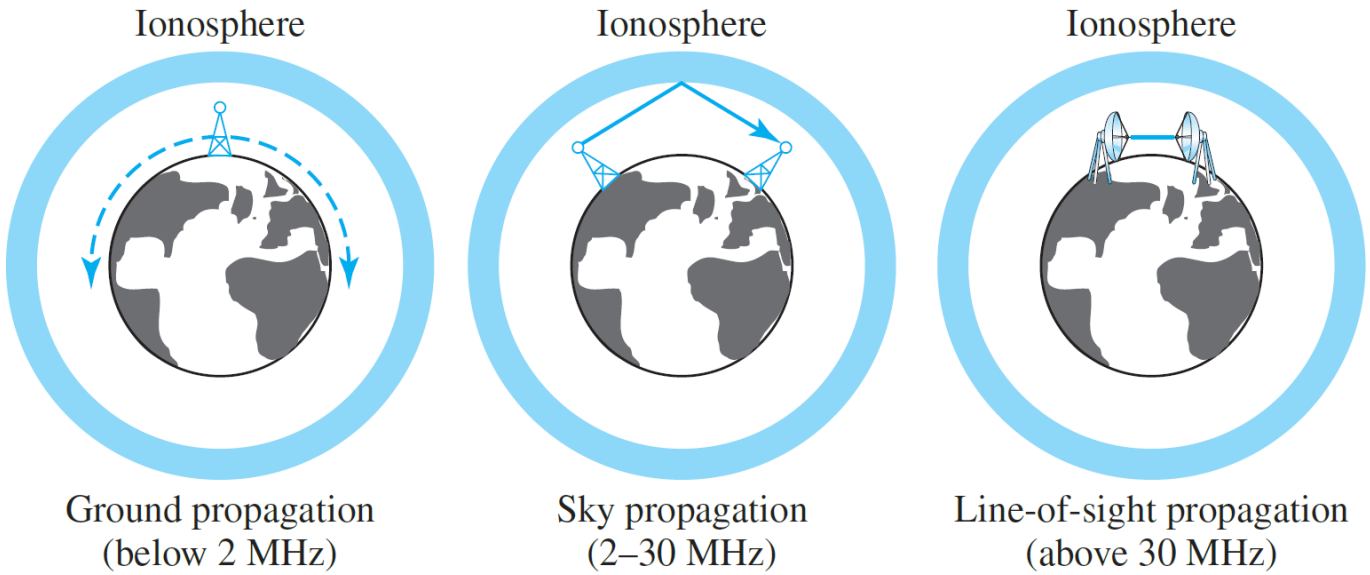
- As we mentioned in FDM section, transmission can be separated into frequency bands

| *Band* | *Range* | *Propagation* | *Application* |
|---|---|---|---|
| very low frequency (VLF) | 3–30 kHz | Ground | Long-range radio navigation |
| low frequency (LF) | 30–300 kHz | Ground | Radio beacons and navigational locators |
| middle frequency (MF) | 300 kHz–3 MHz | Sky | AM radio |
| high frequency (HF) | 3–30 MHz | Sky | Citizens band (CB), ship/aircraft |
| very high frequency (VHF) | 30–300 MHz | Sky and line-of-sight | VHF TV, FM radio |
| ultrahigh frequency (UHF) | 300 MHz–3 GHz | Line-of-sight | UHF TV, cellular phones, paging, satellite |
| superhigh frequency (SF) | 3–30 GHz | Line-of-sight | Satellite |
| extremely high frequency (EHF) | 30–300 GHz | Line-of-sight | Radar, satellite |

# National Spectrum Allocation



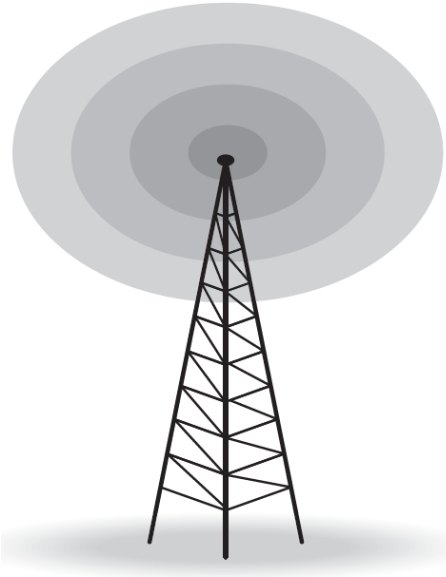
# Wireless Propagation Methods




- Ground propagation
  - **Low** frequency
  - Distance depends on power
- Sky propagation
  - **High**-frequency radio waves radiate upward into the ionosphere where they are reflected back to earth.
  - Greater distance with lower power
- Line-of-sight propagation
  - **Very-high** frequency transmitted in straight lines directly from antenna to antenna (directional)
  - Distance depends on power

# Radio Waves

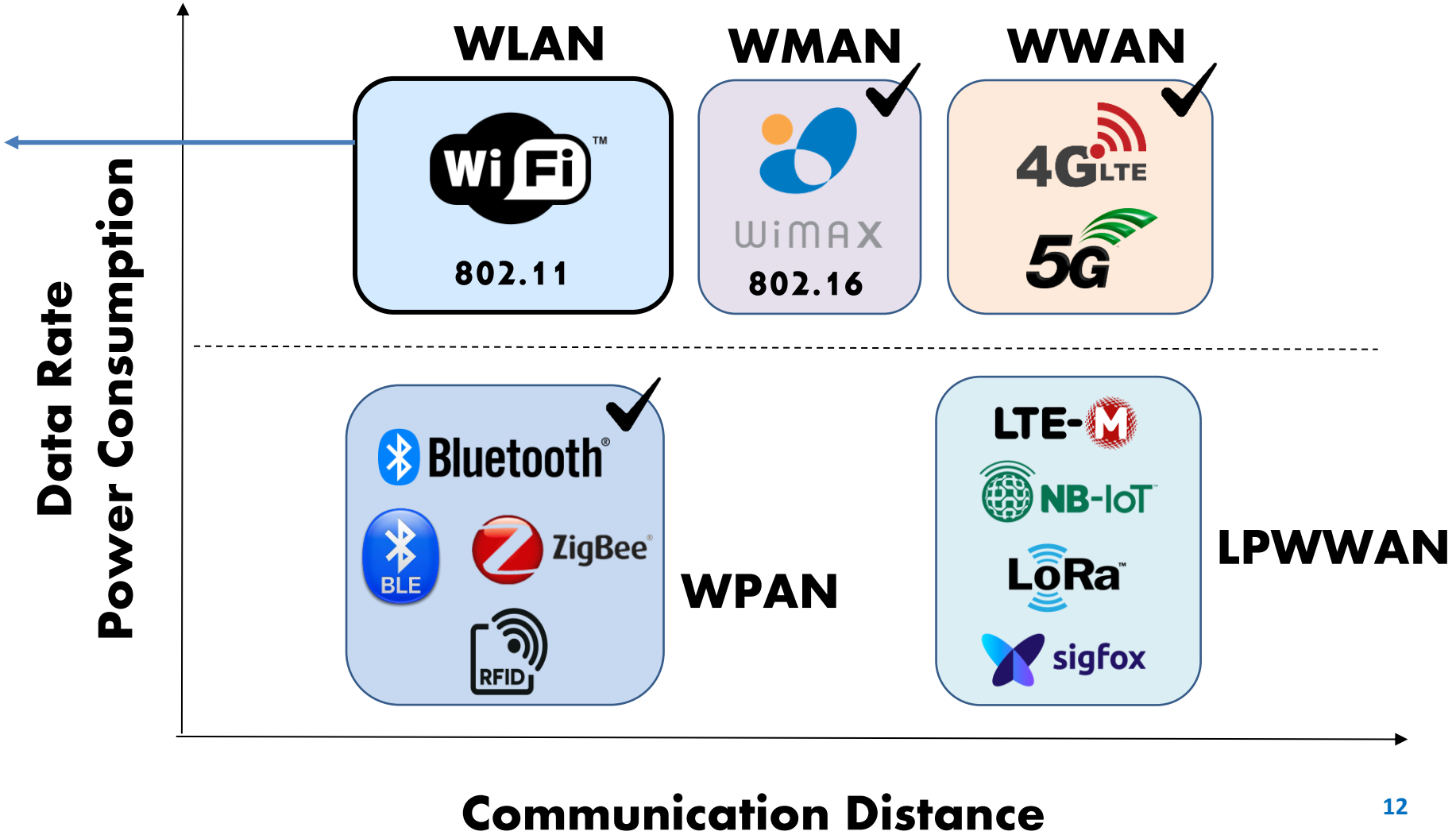
- Frequencies between 3 kHz and 3 GHz
- **Omnidirectional** and propagate in sky mode
- Commonly used in AM, FM, Wi-Fi
- Easily interfered by noise and other signals
- Signals can travel very far and can penetrate walls



# Wireless Technologies

Characteristics:

- Medium Range (Local Area Network)
- High network capacity



# 802.11 vs Wi-Fi

- 802.11 is a standard defined by IEEE
- Wi-Fi = "Wireless Fidelity" (commercially used term)
  - Fidelity = Compatibility between wireless equipment from different manufacturers
  - Wireless technology based on 802.11 standard
- Wi-Fi Alliance: non-profit organization
  - Promote Wi-Fi technology and certify Wi-Fi products
    - Compatibility with other certified equipment
    - Conformance to 802.11 standard
    - Performance meets the required level

# 802.11 Task Groups: Physical Layer

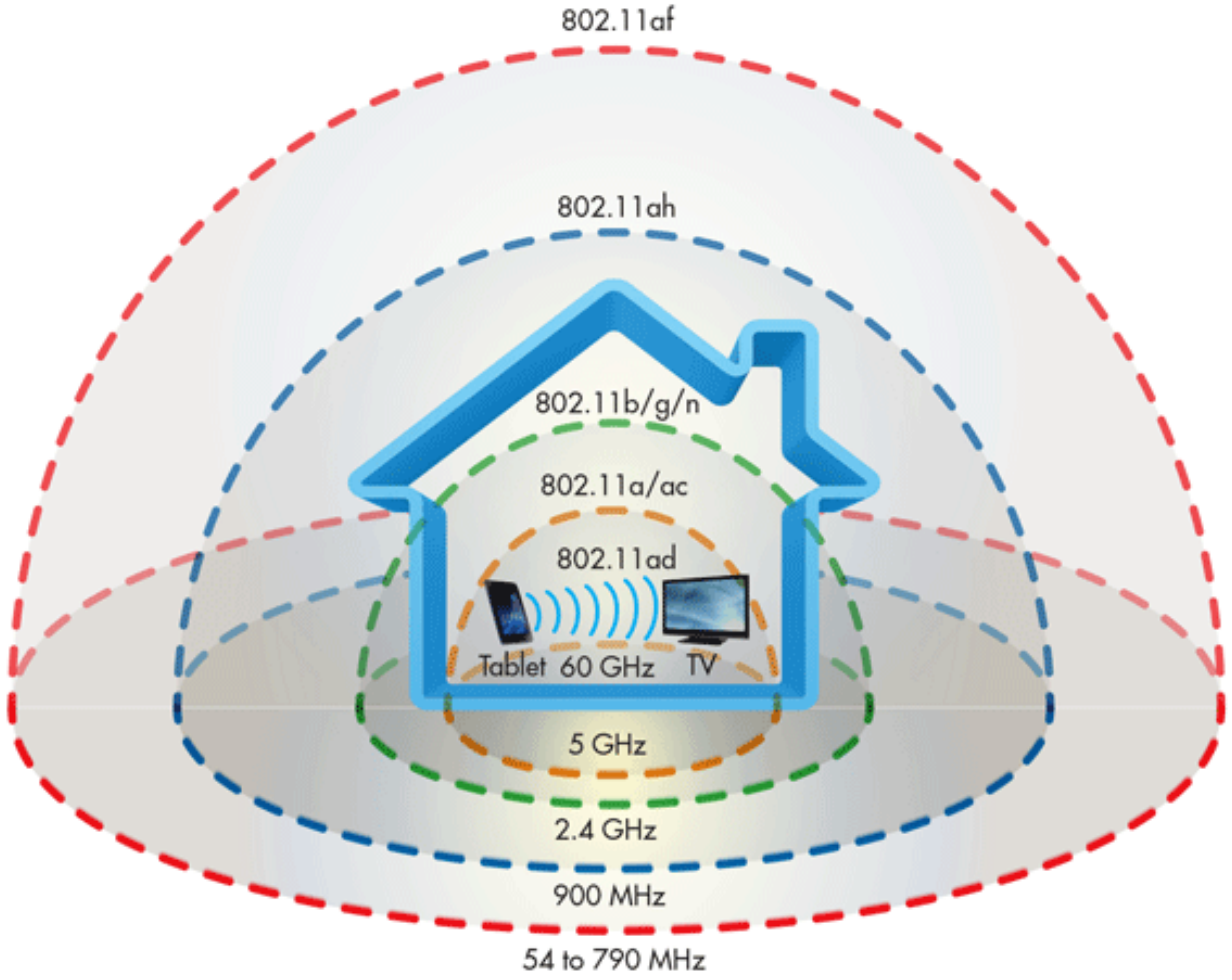
| Task Group | Characteristics |
|---|---|
| 802.11 | Up to 1-2 Mbps in the 2.4 GHz band |
| 802.11b | Up to 11 Mbps in the 2.4 GHz band |
| 802.11g | Up to 54 Mbps in the 2.4 GHz band |
| 802.11a | Up to 54 Mbps in the 5GHz band |
| 802.11n (Wi-Fi 4) | Up to 54-600 Mbps in the 2.4 and 5 GHz bands |
| 802.11ac (Wi-Fi 5) | Up to 6.8 Gbps in the 5 GHz band |
| 802.11ax (Wi-Fi 6) | Up to 11 Gbps in 2.4 and 5 GHz band |
| 802.11be (Wi-Fi 7) | Up to 40 Gbps in 2.4, 5, 6 GHz band |

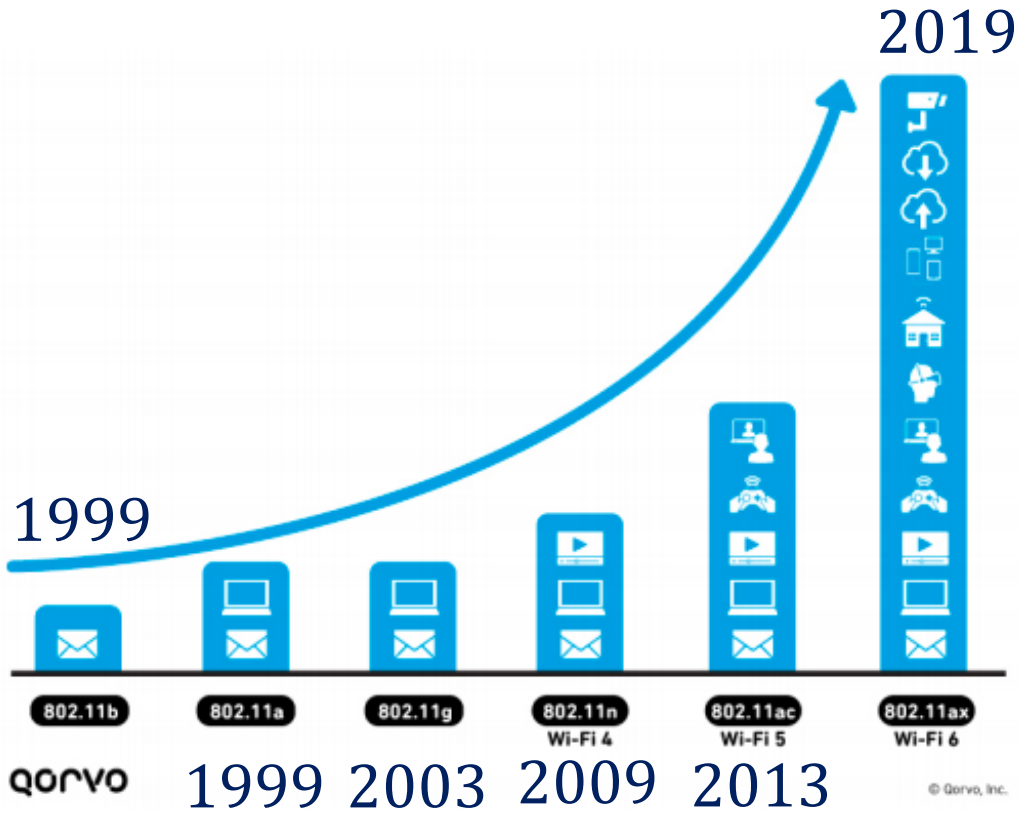
# 802.11 Other Task Groups

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| Task Group | Characteristics |
|---|---|
| 802.11e | MAC enhancement for QoS |
| 802.11i | MAC enhancement for enhanced security |
| 802.11r | Fast roaming |
| 802.11s | Mesh networking |
| 802.11ad (WiGig) | Enable high throughput at 60 GHz frequency |
| 802.11af (White-Fi) | Enable WLAN operation in TV white space (54-790 MHz) |
| 802.11ah (WiFi-HaLow) | Enable WLAN operation at sub-1 GHz license-exempt bands |
| 802.11ae | QoS management |

# 802.11 Standards




http://wwarodom.blogspot.com/2016/08/ieee-80211.html

# WiFi Certified Products

802.11n; also supports 802.11a, 802.11b, and 802.11g




802.11n; also supports 802.11b and 802.11g




802.11g; also supports 802.11b




802.11a; also supports 802.11b and 802.11g






# WLAN Frequency Band

- Operate in **unlicensed frequency band**
  - Advantage: low cost, flexible usage
  - Disadvantage: interference (many users sharing the frequency band)
- Unlicensed frequency band:
  - Industrial, Scientific, and Medical **ISM** bands
  - Unlicensed National Information Infrastructure **UNII** band
- Commonly used band
  - **2.4000–2.4835 GHz ISM** (83.5 MHz): common band for wireless technology, i.e., 802.11 b/g/n, Zigbee, Bluetooth, etc.
  - **5.150–5.825 GHz UNII**: lower (5.150 – 5.250 GHz), middle (5.250 – 5.350) and upper (5.725 – 5.825)



# WLAN Devices

- Client device, e.g., laptop, smart phone, smart appliance, printer, etc.
- Access point
- Wireless router
- Wireless repeater
- Wireless bridge



# Wireless Network Interface Card (WNIC)

- Network interface controller (Wireless LAN card) connects to a wireless antenna and the device
- Work on layer 1 and 2 of the OSI model complying with IEEE 802.11
  - data rate and transmit power (in dBm)
- Mode of operations:
  - **Infrastructure**: devices on the network communicate through an access point
  - **Ad-hoc mode (P2P)**: devices on the wireless network connect directly to each other






# WLAN Antenna

- **Omnidirectional antenna**
- Act as a transmitting/receiving device when attached to a high-frequency AC power source → convert AC voltage and current into electromagnetic wave energy, and vice versa



- Two types of antennas:
  - An **internal antenna:** built into most wireless WNICs, access points, and routers
    - Usually use dipole antenna
  - An **external antenna** necessary for more advanced applications
    - Usually use for **site survey**

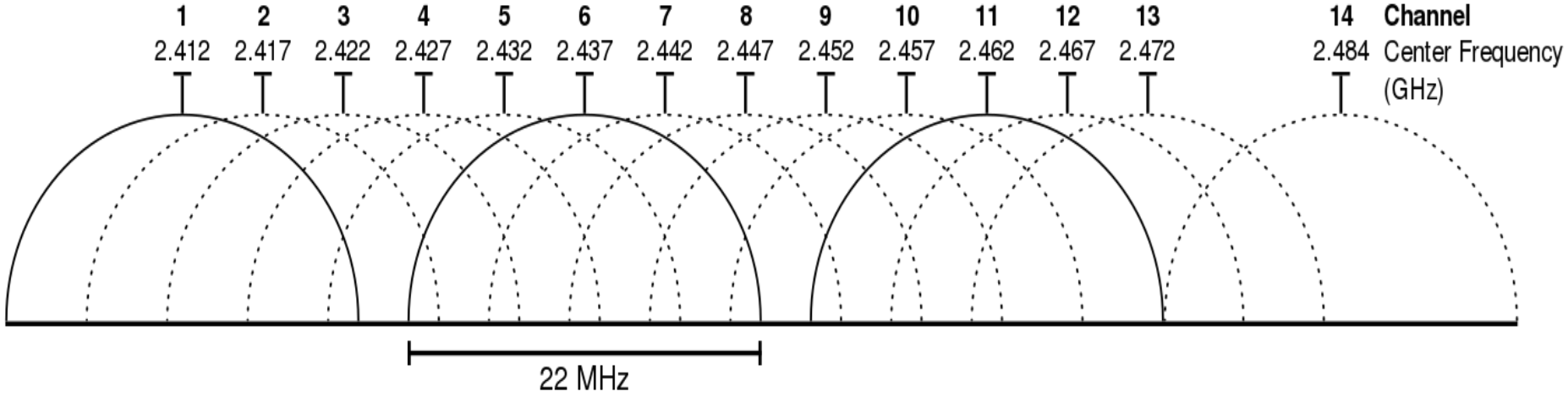


# WLAN Devices: Access Point (AP)



- Multi-radio access point
  - Support two or more WLAN standard simultaneously, e.g. IEEE 802.11 a/b/g
  - Dual-band WiFi supports both 2.4 and 5 GHz frequencies
- **Stealth AP** (hidden access point)
  - Do no broadcast their SSID for security reasons
  - Only users who knows the SSID may connect
  - **SSID:** the Service Set Identifier used to identify the network

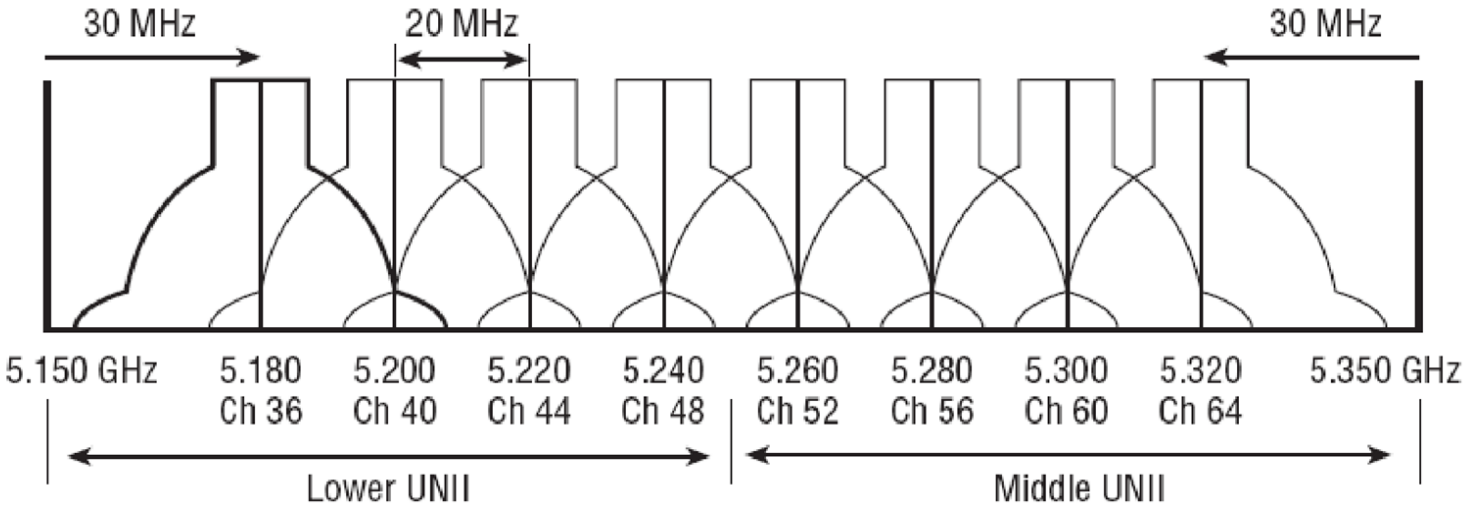
# 802.11b



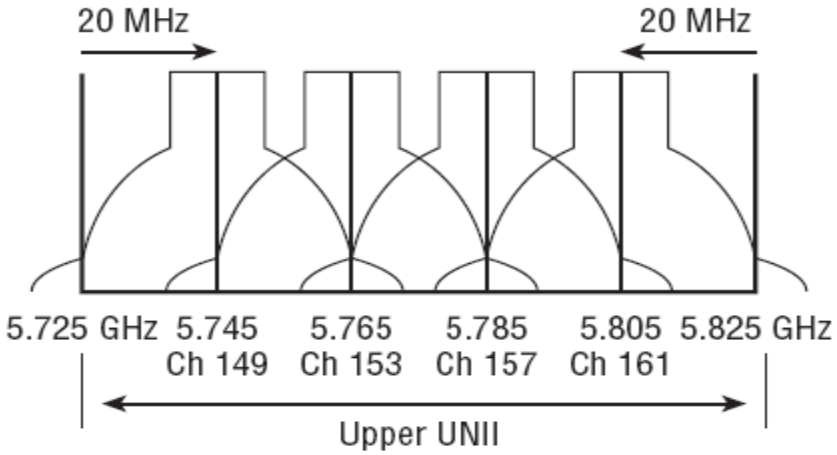

- Operate on the **2.4 GHz band**
  - Up to 14 channels, each is 22 MHz
  - **Channel 1, 6, 11** are not overlapped
- Data rate up to 11 Mbps
- Modulation scheme: QPSK and **CCK**

# 5 GHz UNII Band

**Indoor**

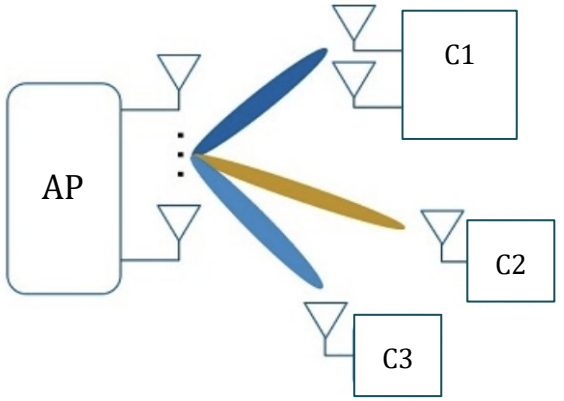



**Outdoor**



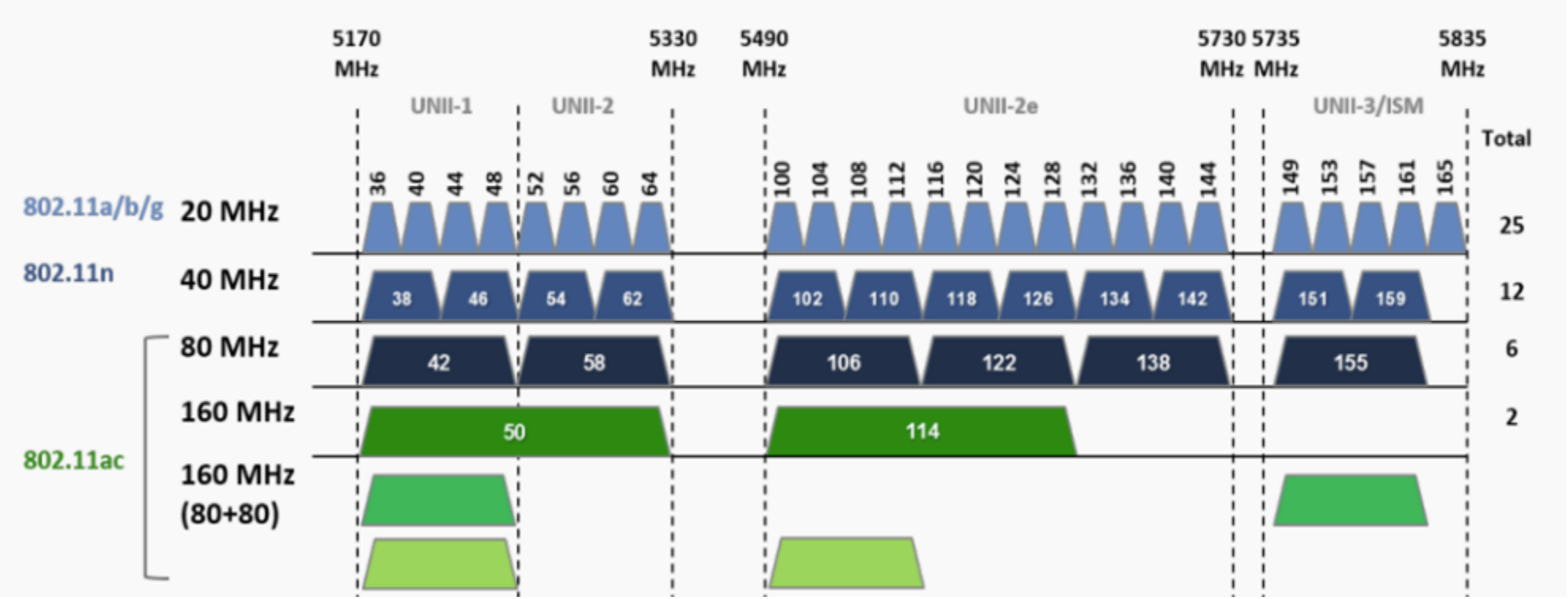
# 802.11ac (Wi-Fi 5)

MU-MIMO




- Operate on the **5 GHz** RF band
- Increase in data rate up to **6.8 Gbps**
- Channel width is 20 / 40 / 80 / **160 MHz** (extended channel bonding)
- Support multiple antenna systems -- **8x8:8 MIMO**
  - **MU-MIMO** (Multi-User MIMO) for downlink communication
  - Up to **4 MU-MIMO clients**
- Transmission scheme: OFDM
- Modulation technique: BPSK, QPSK, QAM (max = **256-QAM**)

# 802.11ac: Channel Bonding




## Up to 160 MHz Channel bonding

http://www.mirazon.com/whats-802-11ac-keep-hearing/

# 802.11ax (Wi-Fi 6)

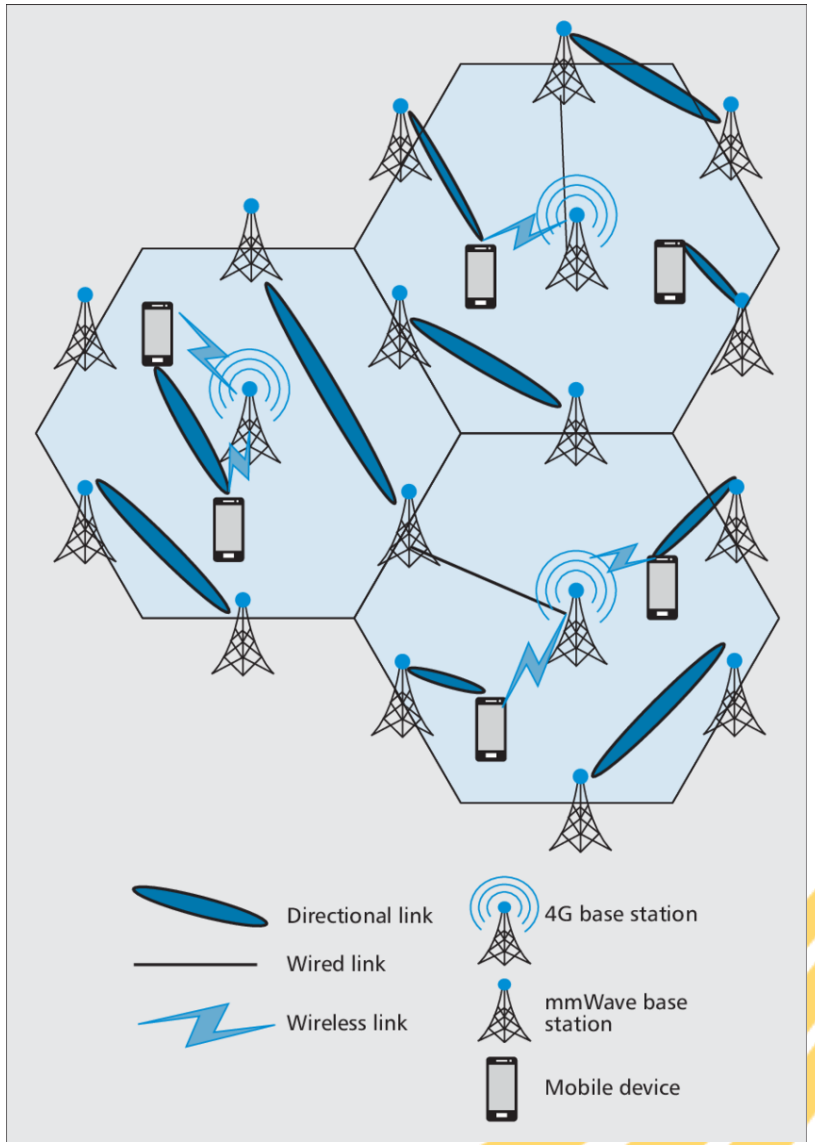
- Operate in both 2.4 and 5 GHz (design to operate in 1-6 GHz ISM band)
- Data rate up to 10 Gbps (but ~4x higher throughput than 802.11ac)
- Channel width is 20 / 40 / 80 / 160 MHz
- Support multiple antenna systems -- **8x8:8 MIMO**
  - **MU-MIMO** (Multi-User MIMO) for both downlink and uplink communication
  - Up to **8 MU-MIMO clients**
- Transmission scheme: **OFDMA**
- Modulation technique: BPSK, QPSK, QAM (max = **1024-QAM**)

# Cellular Site

- Cellular service uses signals in microwave band (700 MHz – 3 GHz mostly).
- **Due to its high frequency, the signals do not travel far.**
- Communication towers(**Base Station**) must be placed efficiently to cover a larger area.



Cell = Hexagon area



# 4G(LTE)/5G Network Frequency Bands in Thailand

4G frequency bands

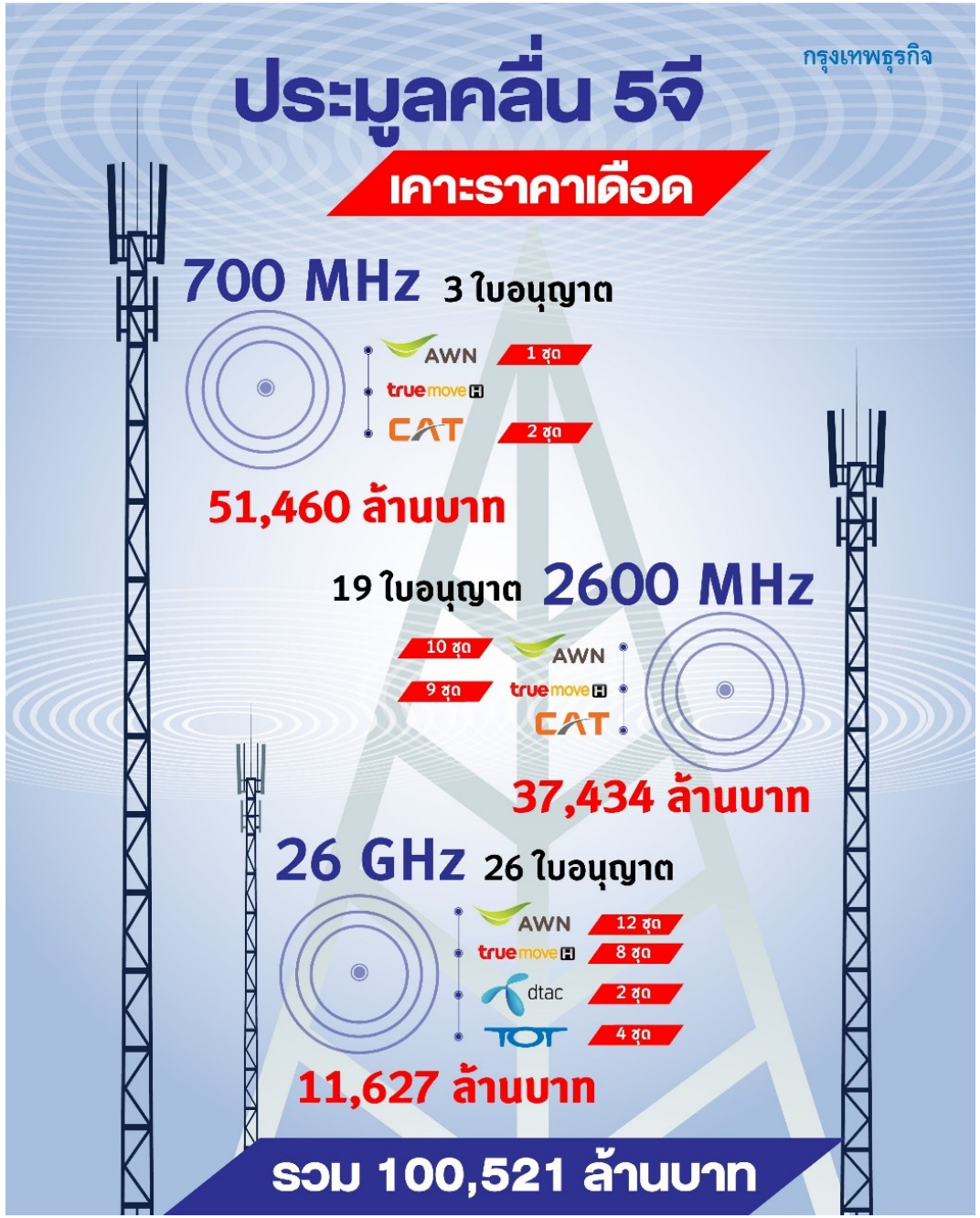
| Frequency Band | 3GPP Band | Operator(s) |
|---|---|---|
| **700 MHz** | Band 28 | AIS, True Corporation |
| **900 MHz** | Band 8 | AIS, True Corporation |
| **1800 MHz** | Band 3 | AIS, True Corporation |
| **2100 MHz** | Band 1 | AIS, True Corporation |
| **2300 MHz** | Band 40 | True Corporation (in partnership with NT) |

5G frequency bands

| Frequency Band | 3GPP Band | | |
|---|---|---|---|
| **700 MHz** | n28 | | |
| **2600 MHz (2.6 GHz)** | n41 | AIS, True Corporation | A key mid-band frequency providing a blend of good coverage and high capacity, crucial for urban and suburban areas. |
| **26 GHz** | n258 | AIS, True Corporation | A high-frequency millimeter-wave (mmWave) band used for delivering ultra-fast speeds in high-density areas such as city centers and major event venues. |

# 1G to 5G Cellular Technology

| Generation | Technology | Speed (Approx.) | Multiplexing | Modulation / Shift Keying |
|---|---|---|---|---|
| 1G | Analog | ~2 Kbps | FDMA | Analog FM |
| 2G | Digital (GSM, CDMA) | ~64 Kbps | TDMA (GSM), CDMA | GMSK (GSM), QPSK (CDMA) |
| 3G | UMTS, CDMA2000 | ~2 Mbps | WCDMA / CDMA2000 | QPSK, 16-QAM |
| 4G | LTE | 100 Mbps – 1 Gbps | OFDMA (DL), SC-FDMA (UL) | QPSK, 16/64-QAM |
| 5G | 5G NR | 1–20 Gbps | OFDMA, Massive MIMO | QPSK, 16/64/256-QAM |
| 6G (~2030) | Terahertz, AI-driven | 100 Gbps – 1 Tbps | OFDMA + new (NOMA, OTFS) | Higher-order QAM, new schemes |




# 5G Auction in Thailand

700 MHz: total 51,460 million THB

2600 MHz: total 37,434 million THB

26 GHz: total 11,627 million THB

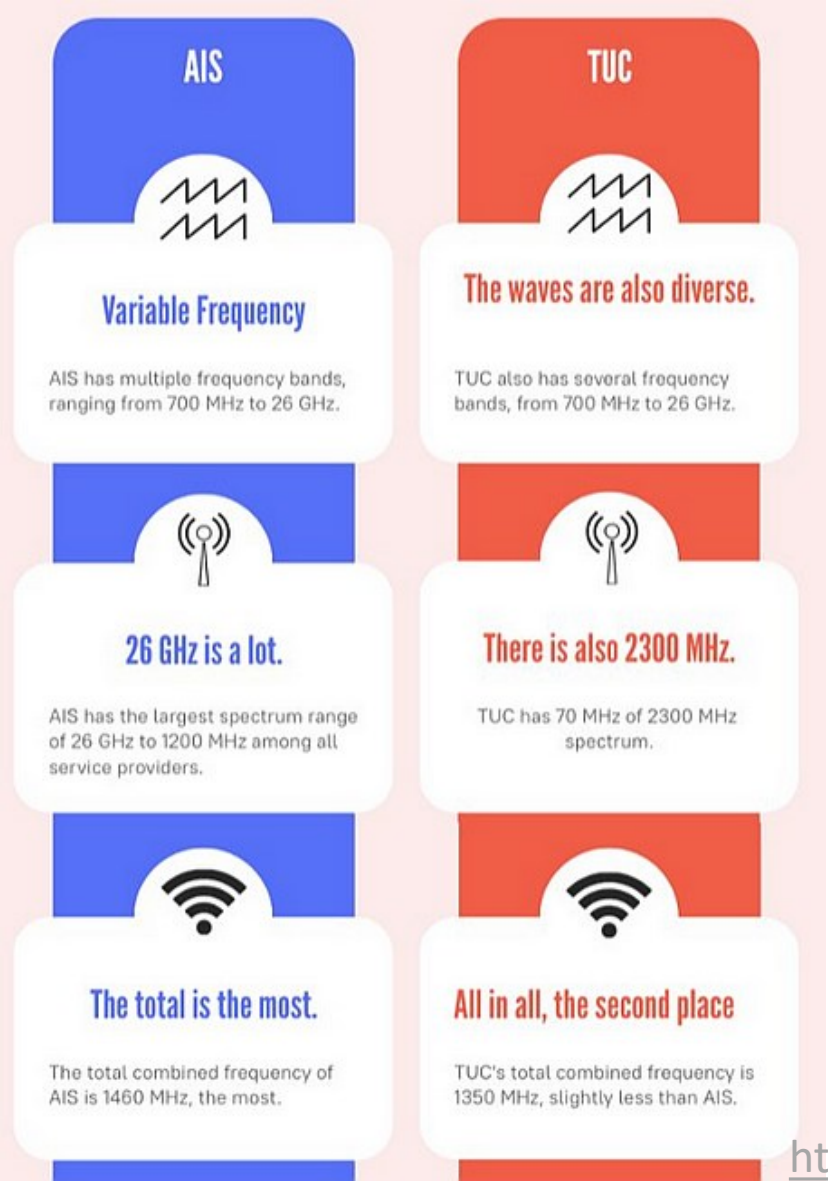
16 Feb 2020

# 5G in Thailand




From 2021 CAT + TOT = NT = National Telecommunication

# Cellular Frequency of AIS vs True




## AIS

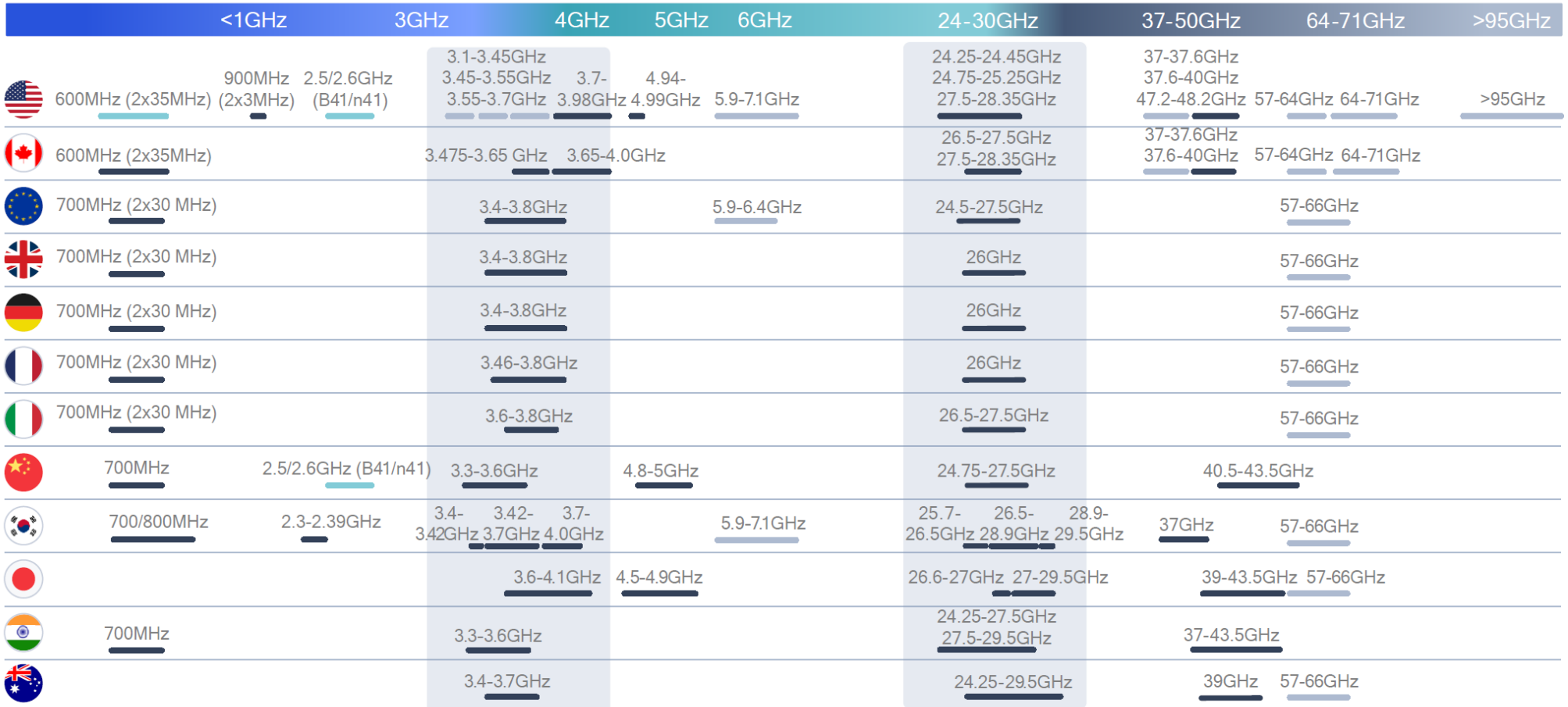
- **Low-Bands (Coverage):** 40 MHz of 700 MHz and 20 MHz of 900 MHz.
- **Mid-Bands (Capacity & Speed):** 40 MHz of 1800 MHz, a newly expanded 90 MHz of 2100 MHz, and a massive 100 MHz block of 2600 MHz (TDD).
- **High-Bands (Ultra-High Capacity):** 1200 MHz of 26 GHz (mmWave).
- **New Total Spectrum: 1490 MHz**

## True Corporation

- **Low-Bands (Coverage):** 40 MHz of 700 MHz and 30 MHz of 900 MHz.
- **Mid-Bands (Capacity & Speed):** 20 MHz of 1500 MHz (SDL), 40 MHz of 1800 MHz, 60 MHz of 2100 MHz, 70 MHz of 2300 MHz (TDD), and 90 MHz of 2600 MHz (TDD).
- **High-Bands (Ultra-High Capacity):** 1000 MHz of 26 GHz (mmWave).
- **Total Spectrum: 1350 MHz**

https://thereporter.asia/eng/2025/06/thailand-spectrum-auction-ais-true/

# Global 5G Spectrum




## Global snapshot of allocated/targeted 5G spectrum

5G is being designed for diverse spectrum types/bands

New 5G band

Licensed

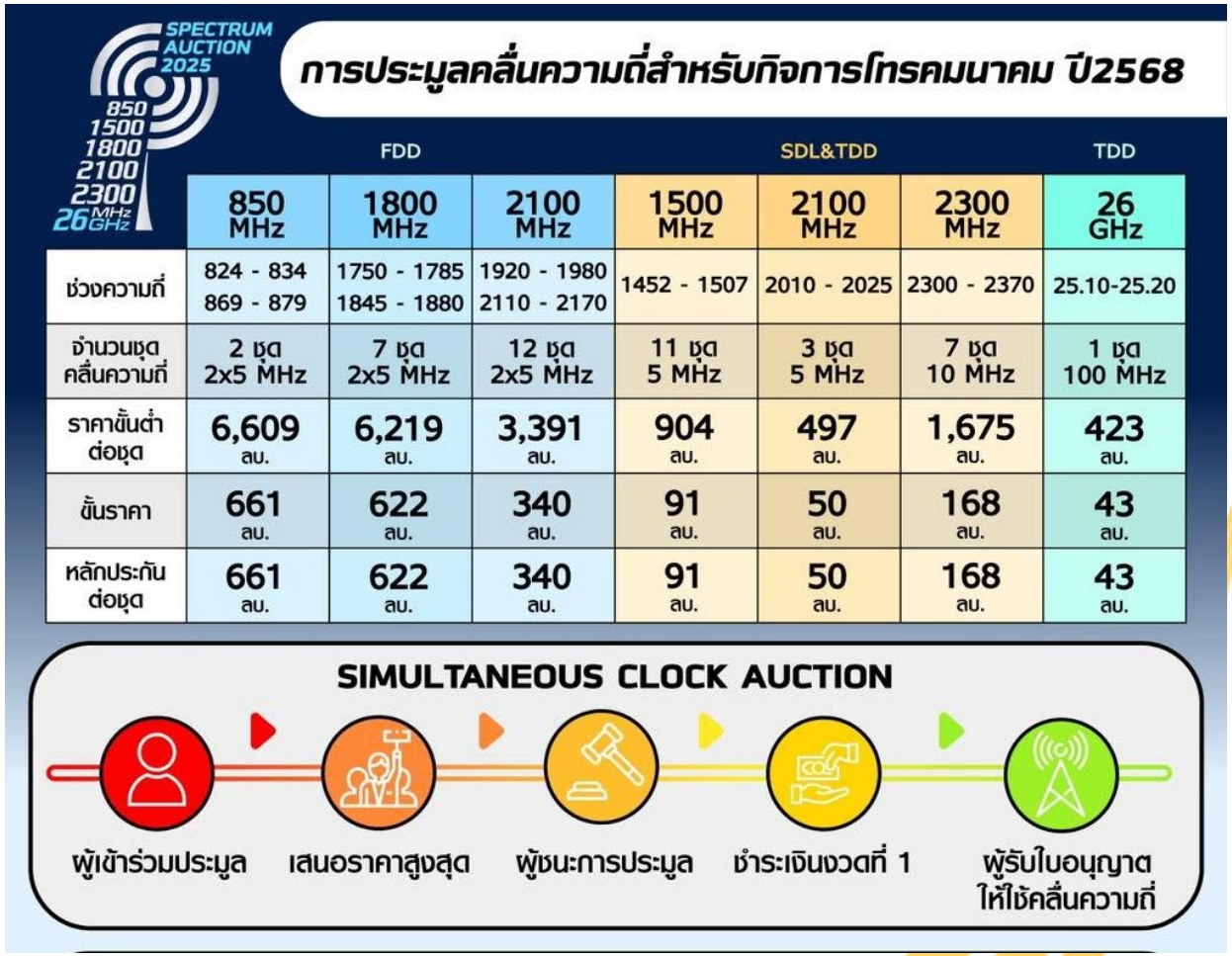
Unlicensed / shared

Existing band

Worldwide allocation of 5G spectrum as on December 2020. Source: Qualcomm 2020a, slide 2.

# Frequency Auction in 2025




https://inews.bangkokbiznews.com/read/486148

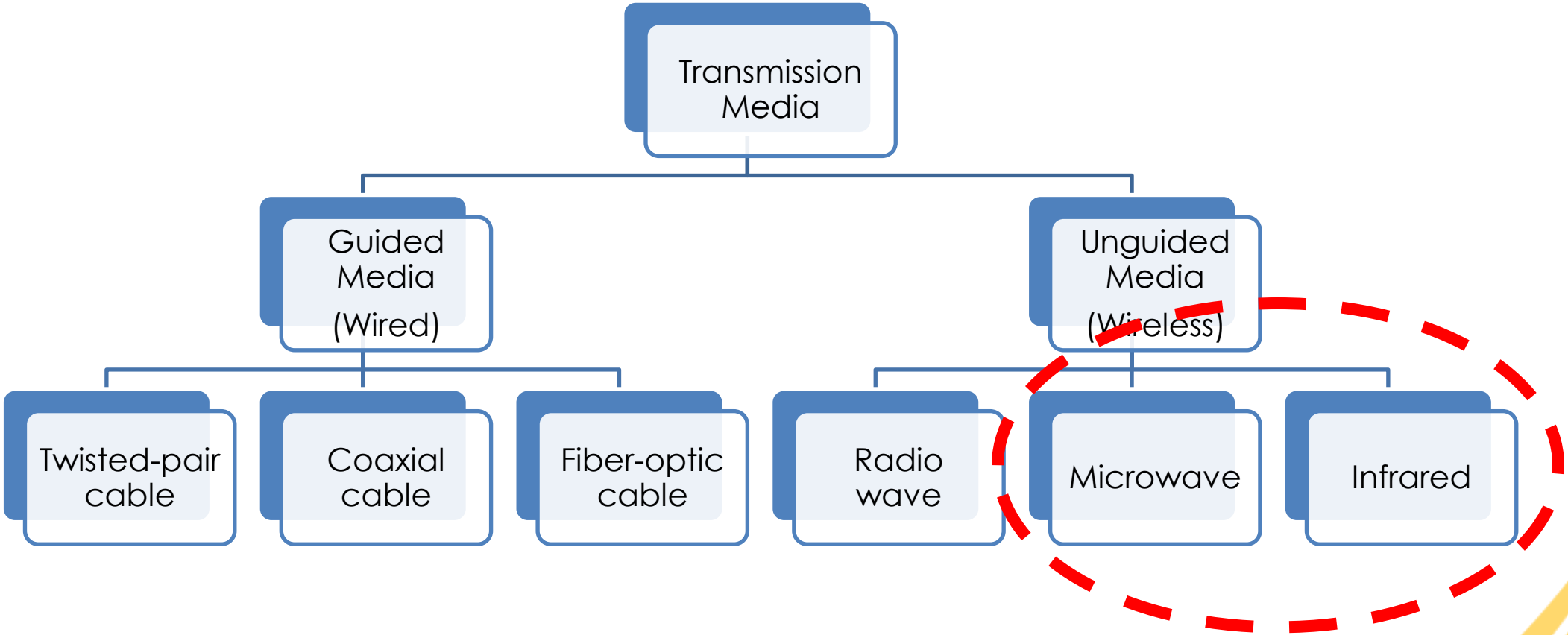
NBTC Facebook https://www.facebook.com/share/p/1DAokTLZcA/




# Auction Results (2025)

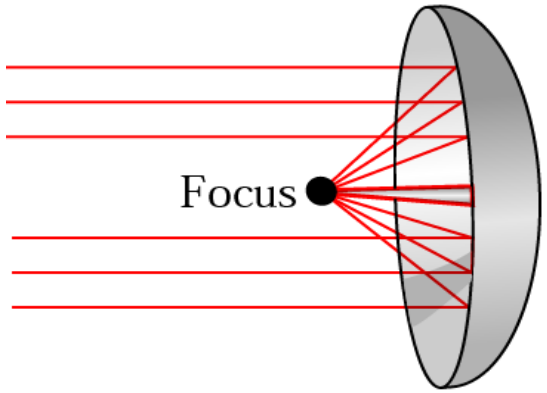
NBTC's Facebook
https://www.facebook.com/photo/?fbid=1134354515384628&set=pcb.1134354538717959

# Transmission Media



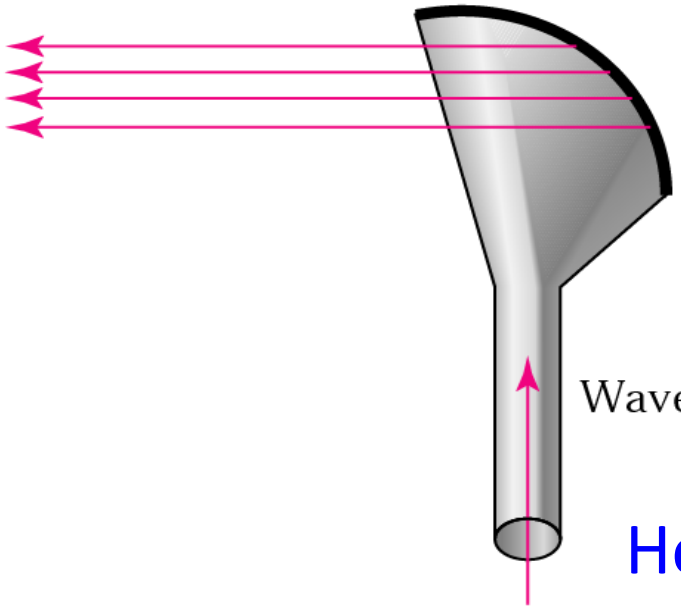
# Microwaves

- Frequencies between 3GHz and 300 GHz
- **Line-of-sight unidirectional**. The sending and receiving antennas need to be aligned. A pair of antennas can be aligned without interfering with another pair of aligned antennas.
- **Very high-frequency** microwaves **cannot penetrate** walls.




Parabolic dish antenna

a. Dish antenna

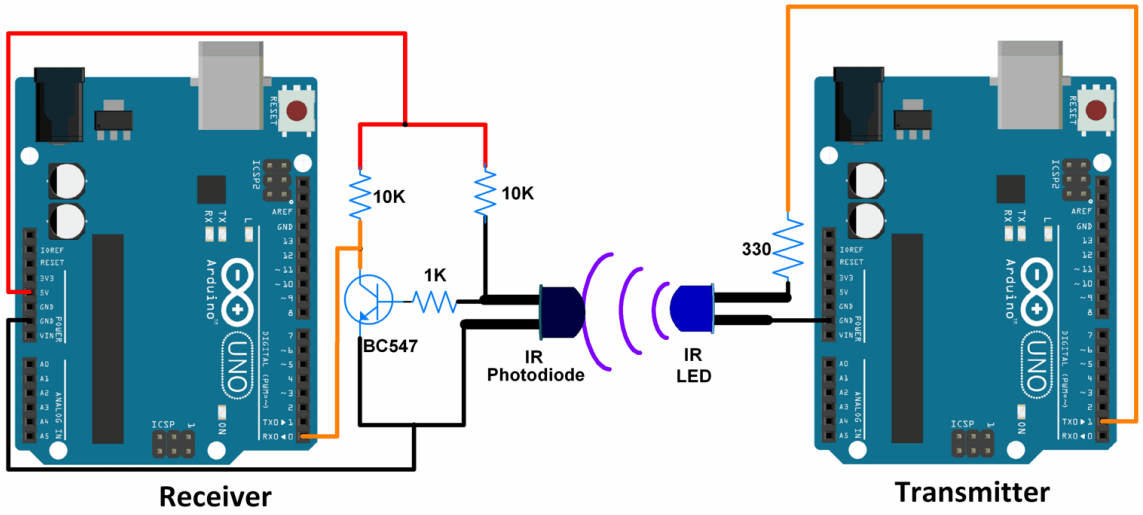


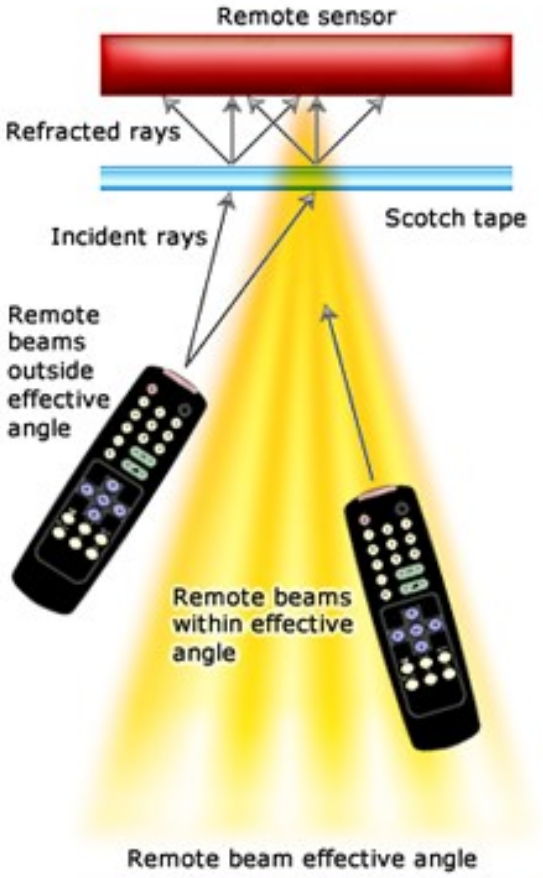

Horn antenna

b. Horn antenna

# Infrared

- Frequencies from 300 GHz to 400 THz
- Infrared waves, having high frequencies, **cannot penetrate walls.**
- Line-of-sight and unidirectional transmission.
- Used for short-range communication.



# How to cable crimping?

- Animation: Ethernet Cables, UTP vs STP, Straight vs Crossover, CAT 5,5e,6,7,8 Network Cables (5.25 mins)
- https://www.youtube.com/watch?v=_NX99ad2FUA

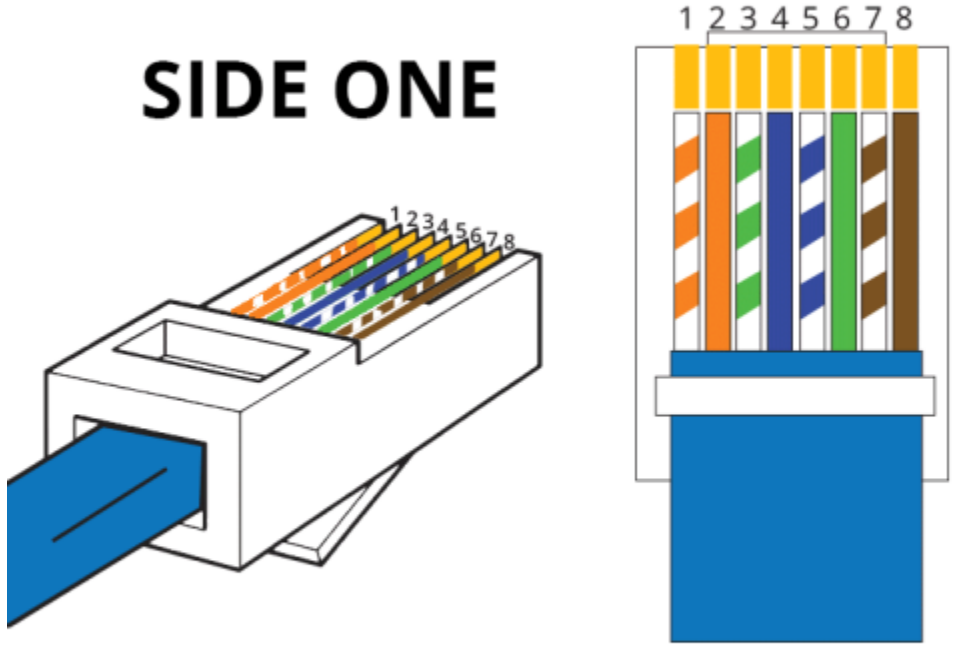
- How to make CAT-5 Cable / Network Wire - Tutorial Guide (4 mins)
- https://www.youtube.com/watch?v=UFlqNQsjYCs

- How To Make RJ45 Network Patch Cables - Cat 5E and Cat 6 (12 mins)
- https://www.youtube.com/watch?v=lullzS740wI

# Connecting RJ-45 Connector to Cable

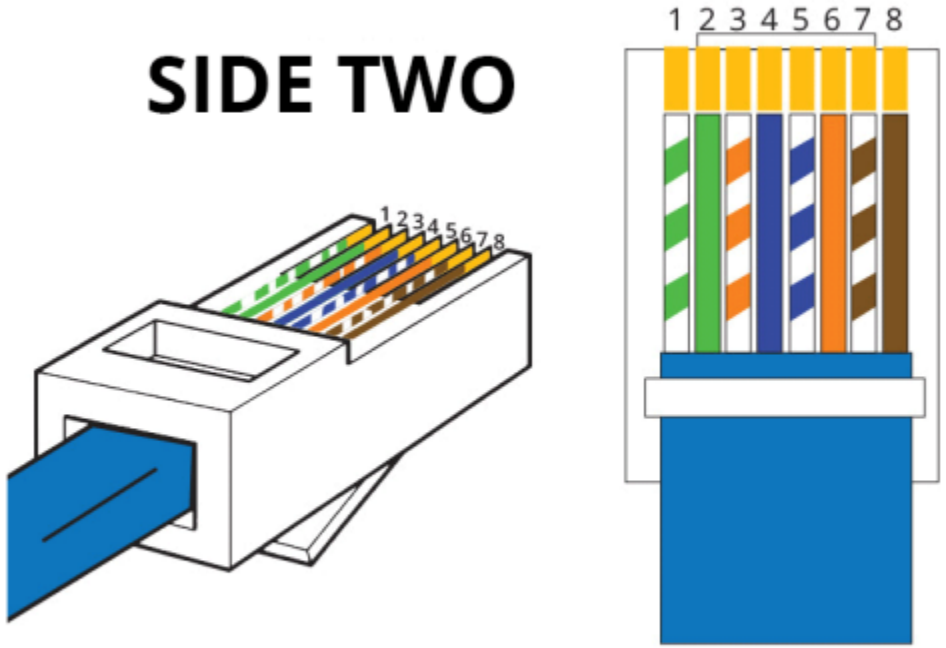
## CROSSOVER PINOUT

### SIDE ONE




| | |
|---|---|
| 1. White Orange | 5. White Blue |
| 2. Orange | 6. Green |
| 3. White Green | 7. White Brown |
| 4. Blue | 8. Brown |

### SIDE TWO




| | |
|---|---|
| 1. White Green | 5. White Blue |
| 2. Green | 6. Orange |
| 3. White Orange | 7. White Brown |
| 4. Blue | 8. Brown |

# Let's Have a Break and

# Do the Cable Crimping lab !







# About the Midterm Exam 2026

- Closed book
- Only One A4 paper is allowed
- Windows' calculator program is allowed
- A scratch paper will be given
  - Write down your name and ID and return it back at the end of the exam



# Reference

Dolvara Gunatilaka, “Lecture 11 Introduction to WLAN and Wi-Fi,” ITCS424: Wireless and Mobile Computing

## Special Thanks


- Special thanks to www.flaticon.com and www.irasutoya.com for free images.